



Sunbeam Institute of Information Technology
Pune and Karad

Module - Operating System Concepts and Linux Programming

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File - types, permissions, groups

type → **-** rwXrwxr-x **1** sunbeam sunbeam 16753 Mar 29 05:49 sample.txt
 permissions/mode/access user/owner group size (bytes) timestamp name

Types of file :

1. Regular file (-)
2. Directory file (d)
3. Link file (l)
4. Pipe file (p)
5. Socket file (s)
6. Char special file (c)
7. Block special file (b)

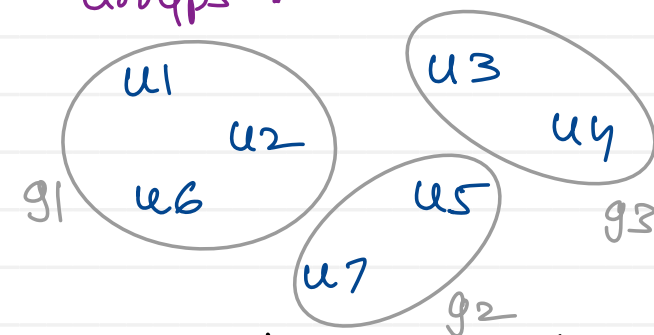
permissions :

r - read
 w - write
 x - execute

levels :

rwX rwX r-x
 ↑ ↑ ↑
 user/ group others
 owner

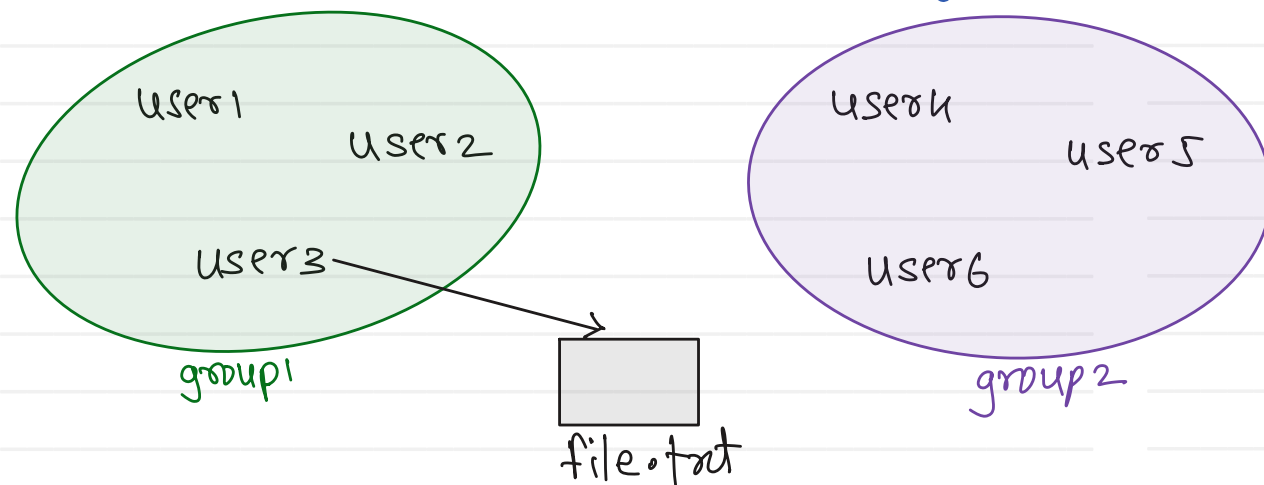
Groups :



u1 has created sample.txt
 Name: sample.txt

use/owner: u1 (rwX)
 group : g1 (u2, u6) (rwX)
 others : g2 & g3 (u3, u4, u5, u7) (r-x)

permission / mode / access : $\underline{rW-}$ $\underline{rW-}$ $\underline{r--}$
 $\downarrow$ $\downarrow$ $\downarrow$
 user/owner group others



($rW-$) owner - users3
 ($rW-$) group - group1 (users1 & users2)
 ($r--$) others - group2 - (users4, users5 & users6)

human format: $rW-$ $rW-$ $r--$
 Numeric format: $\frac{110}{6}$ $\frac{110}{6}$ $\frac{100}{4}$
 (66h)

- add execute for user
- remove write of group

existing : $rW-$ $rW-$ $r--$
 new : rWX $r--$ $r--$
 $\frac{111}{7}$ $\frac{100}{4}$ $\frac{100}{4}$
 (74h) ←

file - file.txt

permissions = rw- rw- r--

user/owner = rw-

group = rw-

others = r--

chmod $\pm$ r/w/x filename

+ - add

- - remove

chmod +x file.txt

chmod -x file.txt

chmod u/g/o $\pm$ r/w/x filename

u - user/owner

g - group

o - others

chmod u+w file.txt

chmod g+r file.txt

chmod u+rwx file.txt

0 - 000

1 - 001

2 - 010

3 - 011

4 - 100

5 - 101

6 - 110

7 - 111

rw-

110

6

rw-

110

6

r--

100

4

(664)

rw-

111

7

rw-

100

4

r--

100

4

(744)

chmod mode filename